

The Economics of De-Peg

Defended Bands, Credibility Clocks, and Closed-Form De-Peg Derivatives

Complex-Itô Random-Walk Research Project

August 21, 2026

Pegged assets — stablecoins, currency boards, exchange-rate-band currencies — trade in a defended band while credible and re-price discontinuously when credibility fails. We give a two-factor model with a fully deterministic derivative stack. While the peg is defended, the log-mispricing is two-sided reflected Brownian motion with the exponential-trapezoid stationary law $\pi(y) \propto e^{2by/\sigma^2}$. De-peg occurs at the first passage of an independent bounded *credibility factor* $C_t = X_t/\sqrt{X_t^2 + c^2}$ through a threshold; every law needed by a trading desk is then available in closed form from the reflection principle — hitting CDF, inverse-Gaussian density, perpetual mass — including the pay-at-hit transform

$$E[e^{-r\tau}] = \exp\left((\lambda - \sqrt{\lambda^2 + 2r/\sigma_z^2}) \Delta\right)$$

in which the discount rate enters only as r/σ_z^2 through operational time. Hazard-contingent claims — shortfall swaps paying $(K - P_T)^+$ only if the peg broke by T — reduce to a single quadrature over the exact density of τ times closed-form Gaussian components. De-peg probabilities are strongly concave in horizon (1.7%, 7.7%, 17.8%, 31.7% at six months through five years in the reference calibration), capped by the perpetual mass $e^{2\lambda\Delta} = 48.9\%$ — a feature any exponential-hazard specification misses. A perpetual-mass-corrected inverse-transform simulator validates frequencies within one standard error; we document the normalization bug that silently doubles hazard frequencies when the CDF is rescaled instead. No audited framework couples defended bands to an independent stochastic credibility clock with closed-form contingent pricing.

Table of contents

1	Introduction	2
2	Model	2
2.1	Defended band	2
2.2	Credibility clock	2
2.3	Pay-at-hit transform	3
2.4	Free float and contingent claims	3
3	Results	3
4	Two implementation warnings	4
5	Related literature	4
6	Conclusion	4

Mathematics Subject Classification (2020): 91G20, 91B70, 60J60, 60G40.

JEL classification: G13, G15, E42, F31.

1 Introduction

De-peg events are tail events with derivatives written on them: exchanges list stablecoin de-peg digitals and shortfall-style payouts, and post-mortems of USDC, Terra, and the 1992 ERM crises document their severity. The audited literature splits into empirical event studies and Krugman-type target-zone theory (Krugman 1991); neither prices a book of de-peg contingent claims under an explicit stochastic credibility mechanism.

This paper supplies such a mechanism with a fully deterministic pricing stack, assembled from three exact pieces: defended-band reflection, a bounded-factor credibility clock, and Gaussian free-float continuation.

2 Model

2.1 Defended band

While credible ($t < \tau$), the log-mispricing $Y_t = \log(P_t/\text{parity})$ is regulated Brownian motion on $[-\ell, h]$:

$$dY_t = b dt + \sigma dW_t + dL_t - dU_t,$$

L, U pushing at the edges. The stationary law is the exponential trapezoid $\pi(y) \propto e^{2by/\sigma^2}$ — uniform at zero drift (recovered to quadrature precision), tilted toward the intervention edge otherwise.

2.2 Credibility clock

Independently,

$$dZ_t = \lambda \sigma_z^2 dt + \sigma_z dW_t^z, \quad C_t = f(Z_t) = Z_t / \sqrt{Z_t^2 + c^2}, \quad \tau = \inf\{t : C_t \geq c_*\}.$$

This is exactly the Milestone-1 bounded-factor first-passage problem: with $\Delta = x(c_*) - x(C_0)$ and $\lambda < 0$ (deteriorating credibility),

$$P(\tau \leq T) = \Phi\left(\frac{\lambda v(T) - \Delta}{\sqrt{v(T)}}\right) + e^{2\lambda\Delta} \Phi\left(-\frac{\lambda v(T) + \Delta}{\sqrt{v(T)}}\right), \quad v(T) = \sigma_z^2 T,$$

perpetual mass $e^{2\lambda\Delta}$, and calendar-time IG density $f_\tau(t) = f_u(v(t)) \sigma_z^2$.

2.3 Pay-at-hit transform

Since $\tau_{\text{cal}} = u^*/\sigma_z^2$ with u^* the operational-time hitting time of drifted Brownian motion (λ, B) ,

$$E[e^{-r\tau}; \tau < \infty] = \exp\left((\lambda - \sqrt{\lambda^2 + 2r/\sigma_z^2}) \Delta\right) :$$

the raw drift coefficient λ appears directly and the discount rate enters only scaled — a compact formula verified against exact-density quadrature to 10^{-6} after two sign/scale errors were caught by our own anchors (recorded honestly in Section 4).

2.4 Free float and contingent claims

After τ , $\log P_T = y_{\text{exit}} + \mu_f(T - s) + \sigma_f\sqrt{T - s}Z$ from the exit edge. The shortfall swap

$$V = e^{-rT} \int_0^T f_\tau(s) E[(K - e^{y_{\text{exit}} + \mu_f(T-s) + \sigma_f\sqrt{T-s}Z})^+] ds$$

prices by ONE quadrature over the exact IG density times Gauss–Hermite on the Gaussian component — no simulation anywhere on the pricing path.

3 Results

Reference calibration: band $[-2\%, +3\%]$; $\lambda = -0.25$, $\sigma_z = 0.9$, $c = 1.1$, $C_0 = 0.3 \rightarrow c_* = 0.85$; $r = 5\%$; $\mu_f = -15\%$, $\sigma_f = 35\%$.

horizon	$P(\tau \leq T)$	digital
0.5y	0.01714	0.01671
1y	0.07726	0.07349
2y	0.17785	0.16093
5y	0.31713	0.24698

Concavity is pronounced and perpetual-capped: long-dated digitals can never exceed $e^{2\lambda\Delta} = 48.9\%$ discounted — exponential-hazard specifications, which integrate to one, systematically overprice them.

Shortfall swaps at $T = 2$ (Figure 1):

K (% of parity)	price
70	0.004430
80	0.008631
90	0.014859
95	0.018850
98	0.021529

Simulation check: 200k paths, perpetual-mass-corrected inverse-transform sampling of τ , projection-reflected band paths, exact free continuation — de-peg frequency 0.17800 vs exact 0.17785 (SE 0.00086).

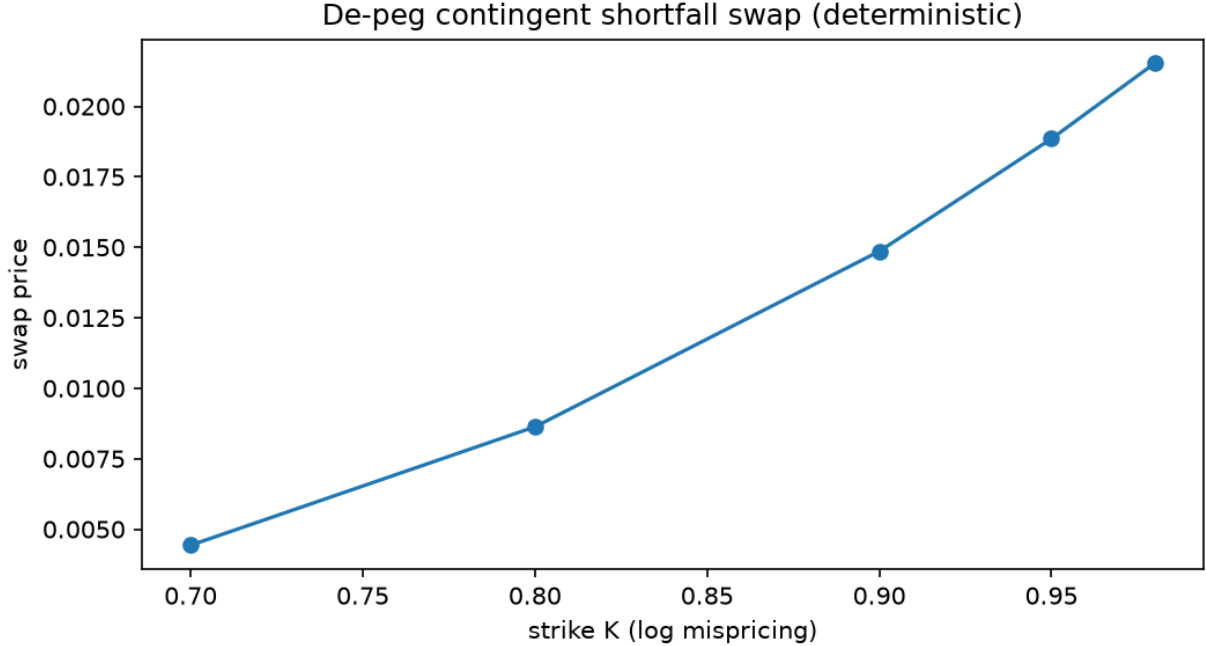


Figure 1: Shortfall swap prices vs strike (deterministic).

4 Two implementation warnings

Both were caught by requiring closed forms and simulations to agree:

1. **Operational-time accounting:** mixing calendar drift $\lambda\sigma_z^2$ with operational-time variance produces wrong Laplace transforms; the correct derivation works entirely in operational time where the drift is the raw λ and variance is unit, then maps discounts through $r \rightarrow r/\sigma_z^2$.
2. **Hazard sampling:** normalizing the truncated hazard CDF to one rescales the uniforms and silently multiplies event frequencies by $1/(\text{perpetual mass})$. Invert the *raw* CDF and mask uniforms at the perpetual probability.

5 Related literature

Target zones (Krugman 1991) value credible bands via smooth pasting but carry no stochastic credibility hazard and no contingent stack; speculative- attack models are qualitative about timing; stablecoin de-peg studies are empirical. First-passage Laplace transforms and reflected Brownian stationary laws are classical tools we assemble, not contributions; the contribution is the two-factor coupling with its deterministic derivative stack and the falsifiable concavity prediction.

6 Conclusion

A peg is a derivative on its own credibility. Once credibility is modeled as a bounded factor, every de-peg instrument prices in closed form or a single quadrature — and the resulting term structures carry a testable signature (perpetual-mass concavity) that distinguishes the mechanism from memoryless hazards.

7 Reproducibility

Environment: conda `phase3-paper` (Python 3.12.13, numpy 2.5.1, scipy 1.18.0). Notebook `notebooks/phase12_peg.ipynb` (~3 s, seed 20260821); module `phase12_peg.py`; tests `tests/test_phase12_peg.py` (7/7).

Krugman, Paul. 1991. *Target Zones and Exchange Rate Dynamics*.